

- 4 (a) Total energy of the oscillating system is the sum of the system's gravitational potential energy (GPE), kinetic energy (KE), and the spring's elastic potential energy (EPE).

Let us analyse the changes in energy as the system goes through one complete oscillation:

At the equilibrium position, the system's KE is maximum. As the mass decelerates upwards, the system's KE and EPE are converted to GPE. At the highest point, the system's GPE is maximum, KE is zero and EPE is minimum.

As the mass accelerates downward towards its equilibrium position, its GPE is converted partly to EPE and partly to KE.

As the mass passes the equilibrium position and continues to decelerate downwards, its KE and GPE are converted to its EPE. At the lowest position, KE is zero, GPE is minimum, and EPE is maximum.

As the mass accelerates upwards towards its equilibrium position, the spring's EPE is converted partly to GPE and partly to KE.

- (b)** An unstable parent nucleus can undergo radioactive decay to emit an alpha particle, beta particle or gamma radiation. In alpha and beta decay, the stationary parent nucleus disintegrates into two parts: a daughter nucleus and either an alpha or beta particle.

The products (daughter nucleus plus either alpha or beta particle) have a total rest-mass energy lower than that of the parent nucleus. The lost mass Δm is converted into energy according to Einstein's mass-energy equivalence equation $E = \Delta mc^2$ where c is the speed of light in free space.

The daughter has a higher binding energy per nucleon and is more stable than the parent nucleus. The combined binding energy of the products is higher than the binding energy of the parent nucleus. The difference in binding energy of the products and parent nucleus is another way to find the nuclear energy released.

Most of the energy is released as the kinetic energy of the products, with the remainder as electromagnetic radiation mainly in the form of gamma radiation.

The gamma radiation comes from the de-excitation of the daughter nucleus formed in alpha or beta decay. The daughter nucleus, which is in an excited state, de-excites and falls to a lower energy state. The energy difference is emitted as gamma radiation (radiation energy).